

Comments in Python

Manjunath Prasad H. R.

January 22, 2025

Some Anonymous Quotes (AI generated) on Comments

1. “Code tells you how; comments tell you why.”
2. “Comments are the breadcrumbs of logic in the forest of code.”
3. “Good code is self-explanatory; great code has comments.”
4. “Write comments as if the next programmer is a violent sociopath who knows where you live.”
5. “Comments are like road signs; ignore them, and you’re lost.”
6. “Code is poetry; comments are its annotations.”
7. “A well-commented line is worth a thousand lines of confusion.”
8. “Comments aren’t excuses for bad code, but explanations for good code.”
9. “Code is for machines; comments are for humans.”
10. “Without comments, code is a puzzle without clues.”

COMMENTS are an extremely useful feature in most programming languages. As your programs become longer and more complicated, you should add notes within your programs that describe your overall approach to the problem you’re solving. A *comment* allows you to write notes in English within your programs[1].

Python ignores comments, and you can use them to write notes or remind yourself what the code is trying to do. Any text for the rest of the line following a hash mark (#) is part of a comment.

Sometimes, programmers will put a # in front of a line of code to temporarily remove it while testing a program. This is called commenting out code, and it can be useful when you’re trying to figure out why a program doesn’t work. You can remove the # later when you are ready to put the line back in.

Python also ignores the blank line after the comment. You can add as many blank lines to your program as you want. This can make your code easier to read, like paragraphs in a book[3].

Comments in Python start with the hash character, #, and extend to the end of the physical line. A comment may appear at the start of a line or following whitespace or code, but not within a string literal. A hash character within a string literal is just a hash character[2].

1 Single-line comments

- Start with the `#` symbol.
- Everything after `#` on that line is treated as a comment and ignored by the interpreter.

```
1 # This is a single-line comment
2 print("Hello, World!") # This prints a message
```

2 Multi-line comments (using multiple `#` symbols)

- For longer comments, you can use multiple single-line comments.

```
1 # This is a multi-line comment
2 # that spans across multiple lines.
3 print("Multi-line comments!")
```

3 Multi-line comments (using triple quotes)

- Triple quotes (`"""` or `"""`) can be used for multi-line comments.
- However, these are technically treated as multi-line strings, not comments. They are ignored by the interpreter only if not assigned to a variable.

```
1 """
2 This is a multi-line comment-like string
3 that spans several lines.
4 """
5 print("Triple-quoted multi-line comment-like string!")
```

4 In-line comments

- Used on the same line as code, usually to explain what the code does.

```
1 x = 10 # Initialize x with 10
2 print(x) # Output the value of x
```

5 Docstrings

- Triple quotes are also used for documentation strings, or docstrings, which are special multi-line comments used to describe a module, class, or function.
- Unlike regular comments, docstrings are accessible using Python's help system (`help()`).

```
1 def add(a, b):
2     """
3     This function takes two numbers and returns their sum.
4     Parameters:
5         a (int): The first number
6         b (int): The second number
```

```
7     Returns:
8         int: The sum of a and b
9     """
10    return a + b
```

References

- [1] Eric Matthes. *Python Crash Course: A Hands-On, Project-Based Introduction to Programming*. No Starch Press, San Francisco, CA, 1st edition, 2016.
- [2] Python Software Foundation. The python programming language. <https://docs.python.org/3/tutorial/introduction.html>, 2025. Accessed: 2025-01-15.
- [3] Al Sweigart. *Automate the Boring Stuff with Python: Practical Programming for Total Beginners*. No Starch Press, 2nd edition, 2019.